

Functional Annotation of *pvdH* (Q88F75 / PP_4223) in *Pseudomonas putida* KT2440

1. Summary / Answer to the Research Question

pvdH encodes **L-2,4-diaminobutyrate:2-oxoglutarate aminotransferase (PvdH; EC 2.6.1.76)**, a soluble, cytoplasmic, pyridoxal-5'-phosphate (PLP)-dependent enzyme of the class-III aminotransferase family. Its primary function is to transaminate **L-aspartate- β -semialdehyde (ASA)** with the amino group of **L-glutamate**, producing **L-2,4-diaminobutyrate (DAB)** and **2-oxoglutarate (α -ketoglutarate)**. DAB is one of the two building blocks—together with D-tyrosine—that are condensed to form the fluorescent dihydroxyquinoline **chromophore of pyoverdine**, the major siderophore of *P. putida* KT2440. PvdH therefore supplies an essential precursor for **pyoverdine-mediated iron acquisition**, acting upstream in the cytoplasm before the pyoverdine peptide is assembled, exported to the periplasm for maturation, and secreted.

2. Gene / Protein Identity (verified)

- **UniProt:** Q88F75 — "Diaminobutyrate-2-oxoglutarate transaminase," EC 2.6.1.76
- **Gene / locus:** *pvdH* / PP_4223
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125)
- **Family / domains:** Class-III PLP-dependent aminotransferase — Aminotrans_3 (IPR005814), Aminotrans_3_PPA_site (IPR049704), **Dat** (IPR004637, diaminobutyrate aminotransferase), PyrddlP-dep_Trfase (IPR015424/IPR015421).

Identity verification: The gene symbol, EC number, and domain architecture are fully consistent. The direct functional characterization of PvdH was performed on the *P. aeruginosa* PAO1 ortholog, but that study explicitly showed that PvdH homologues reside in the pyoverdine-synthesis loci of other pyoverdine-producing *Pseudomonas*, and *P. putida* KT2440 is an established pyoverdine producer. Q88F75 is therefore correctly annotated as the *P. putida* PvdH ortholog. No conflicting "same-symbol, different-gene" literature was encountered.

3. Primary Molecular Function and Substrate Specificity

The definitive biochemistry comes from Vandenende, Vlasschaert & Seah (2004), who subcloned, hyperexpressed, and purified PvdH and characterized it enzymatically (P 15317763):

- **Reaction catalyzed:** aminotransferase interconverting **aspartate- β -semialdehyde $\rightleftharpoons$ L-2,4-diaminobutyrate**, using α -ketoglutarate/L-glutamate as the keto-acid/amino donor pair.
- **Kinetic mechanism:** classic **ping-pong bi-bi**, as expected for PLP transaminases (the PLP cofactor cycles between pyridoxal and pyridoxamine forms).
- **Substrate specificity for the amino acceptor:** **highest specificity for α -ketoglutarate**; specificity toward pyruvate is **41-fold lower**; **negligible activity** toward other keto acids tested. This establishes glutamate/ α -ketoglutarate as the physiological amino shuttle.

"PvdH was found to catalyze an aminotransferase reaction, interconverting aspartate beta-semialdehyde and L-2,4-diaminobutyrate. Steady-state kinetic analysis ... established that the enzyme adopts a ping-pong kinetic mechanism and has the highest specificity for alpha-ketoglutarate."

(P 15317763)

Genetic requirement: *pvdH* (and *asd*, aspartate- β -semialdehyde dehydrogenase, which makes the ASA substrate) knockouts **cannot synthesize pyoverdine under iron limitation unless L-2,4-diaminobutyrate is supplied exogenously**, proving that PvdH is the physiological source of DAB for the chromophore (P 15317763).

Mechanistically, PvdH belongs to the same class-III aminotransferase / **Dat (diaminobutyrate aminotransferase)** family as EctB (the DAB aminotransferase of ectoine biosynthesis) and DABA-AT of other pathways, all of which perform the ASA $\rightarrow$ DAB transamination. The InterPro "Dat" signature (IPR004637) on Q88F75 independently corroborates this assignment by sequence.

3.1 Sequence / structure-level evidence for the target protein itself

Because the enzymatic characterization above was performed on the *P. aeruginosa* PAO1 ortholog, I independently verified that the *P. putida* protein (Q88F75) carries the same functional identity:

- **Length & cofactor:** Q88F75 is a **452-residue** protein annotated with a **pyridoxal-5'-phosphate (PLP)** cofactor (ChEBI:597326), the hallmark of aminotransferases.
- **No export/membrane signals:** the entry has **no signal peptide and no transmembrane segment** (only an N-terminal disordered, polar-biased region, residues 8–27), consistent with a **soluble cytoplasmic** enzyme.
- **Orthology:** a full-length Needleman–Wunsch alignment against the biochemically characterized PAO1 PvdH (Q9I168 / PA2413, 469 aa) gives **83.6 % amino-acid identity (377/451 aligned positions)**. This near-full-length, high identity establishes clear orthology and justifies transferring the experimentally measured ASA⇒DAB aminotransferase activity to Q88F75.

4. Subcellular Localization

PvdH is predicted to be a **soluble cytoplasmic enzyme**. Evidence and inference: - Class-III PLP aminotransferases (Aminotrans_3 fold) are canonically soluble cytoplasmic proteins; Q88F75 has no signal peptide or transmembrane segment consistent with this. - Its product, the amino acid DAB, is a **cytoplasmic metabolite** incorporated by cytoplasmic non-ribosomal peptide synthetases (NRPS) into the pyoverdine precursor (ferribactin). Pyoverdine biosynthesis begins in the cytoplasm; the precursor is subsequently translocated to the **periplasm** for chromophore maturation and then secreted. PvdH acts in the **cytoplasmic (precursor-supply) phase**, upstream of periplasmic maturation.

The downstream compartmentalization is well documented in *P. putida* KT2440: deleting the **PvdRT-OpmQ** and **MdtABC-OpmB** efflux systems causes pyoverdine to accumulate **in the periplasm** ("more pyoverdine in the periplasm,"

P 30346656

and PvdT/PvdR were biochemically shown to directly interact with pyoverdine and hydrolyze ATP

(
P 36807028

These place secretion/maturation in the periplasm/membrane, distinct from PvdH's cytoplasmic role.

4.1 Genomic locus context (synteny) in KT2440

Genomic-neighborhood analysis of the KT2440 chromosome confirms, in the **exact target strain**, that *pvdH* (PP_4223) is **embedded within the pyoverdine biosynthesis gene cluster**, co-localized with the peptide-assembly, export, and uptake machinery:

Locus	Gene	Product
PP_4215	<i>pvdO</i>	Pyoverdine biosynthesis-related protein
PP_4216	<i>pvdE</i>	Pyoverdine ABC export system (fused ATPase/permease)
PP_4217	<i>fpvA</i>	TonB-dependent ferripyoverdine outer-membrane receptor
PP_4219	<i>pvdD</i>	Non-ribosomal peptide synthetase (ferribactin synthase)
PP_4220	<i>pvdJ</i>	NRPS subunit of ferribactin synthase
PP_4221	<i>pvdI</i>	NRPS subunit of ferribactin synthase
PP_4222	—	SyrP-like tailoring protein
PP_4223	pvdH	Diaminobutyrate-2-oxoglutarate transaminase (this study)

This synteny is precisely the pyoverdine-locus co-localization predicted by Vandenende et al. (2004) for PvdH orthologs, and provides independent genomic evidence that PP_4223 acts specifically in pyoverdine biosynthesis (supplying DAB to the adjacent NRPS enzymes) rather than in generic transamination.

5. Pathway Context and Biological Process

PvdH functions in the **pyoverdine siderophore biosynthesis pathway**, the principal high-affinity iron-acquisition system of fluorescent pseudomonads:

1. **Precursor supply (cytoplasm):** Asd generates aspartate- β -semialdehyde; **PvdH transaminates it to L-2,4-diaminobutyrate (DAB)**. In parallel, D-tyrosine is provided.

2. **Chromophore/peptide assembly:** DAB + D-tyrosine are condensed by NRPS machinery, building the pyoverdine peptide backbone and the precursor of the fluorescent dihydroxyquinoline chromophore.
3. **Maturation & export (periplasm/membrane):** the precursor is transported to the periplasm, matured, and secreted via ABC/RND efflux systems (**PvdRT-OpmQ**, with **MdtABC-OpmB** contributing;

P 30346656

P 36807028

4. **Iron capture & uptake:** secreted apo-pyoverdine chelates Fe(III); the ferri-pyoverdine complex is imported through the outer-membrane receptor **FpvA**.

Physiological importance in *P. putida* KT2440: - KT2440 produces a structurally characterized pyoverdine as its **major (essentially sole) siderophore** (

P 19459056

so PvdH-derived DAB is essential for its principal iron-scavenging system under iron limitation. - Pyoverdine iron acquisition also controls **surface motility**: pyoverdine (ppsD) or FpvA-receptor mutants fail to spread on surfaces, and iron availability acts as an environmental cue for swarming-like movement at 18–28 °C (

P 17564617

We deliberately restrict the annotation to this precise precursor-supply role; broader pleiotropic iron-linked phenotypes are mentioned only insofar as they establish the pathway's function.

6. Supported and Refuted Hypotheses

Supported - PvdH is a PLP-dependent class-III aminotransferase (EC 2.6.1.76) catalyzing $ASA \rightleftharpoons DAB$ — supported by direct enzymology (

P 15317763

and domain evidence (IPR004637 "Dat"). - α -Ketoglutarate/glutamate is the physiological amino-shuttle pair — supported by 41-fold specificity over pyruvate (

P 15317763

- PvdH provides the DAB precursor required for pyoverdine — supported by knockout complementation with exogenous DAB
(

P 15317763

- The pathway serves iron acquisition and pyoverdine-dependent surface motility in KT2440
(

P 19459056

).

P 17564617

Refuted / excluded - PvdH is **not** a broad-specificity transaminase: it has negligible activity toward keto acids other than α -ketoglutarate (and, weakly, pyruvate)
(

P 15317763

- PvdH is **not** a membrane/secretion component: export and periplasmic maturation are handled by distinct proteins (PvdRT-OpmQ, MdtABC-OpmB;
(

P 30346656

).

P 36807028

7. Limitations and Future Directions

- The direct enzymatic characterization was performed on the *P. aeruginosa* PAO1 ortholog; the *P. putida* KT2440 protein (Q88F75) itself has not, to our knowledge, been individually purified and kinetically characterized. Given high sequence conservation and shared pyoverdine-locus context, function is confidently inferred but not experimentally demonstrated for KT2440 specifically.
- Subcellular localization of PvdH is inferred from family/fold and pathway logic rather than measured directly (e.g., fractionation or fluorescent tagging).
- A crystal structure of PvdH would confirm the PLP-binding lysine and active-site residues; comparative modeling against EctB/DABA-AT structures could refine substrate-binding determinants.

Key References

- Vandenende CS, Vlasschaert M, Seah SYK. *Functional characterization of an aminotransferase required for pyoverdine siderophore biosynthesis in Pseudomonas aeruginosa PAO1*. J Bacteriol, 2004.

P 15317763

- Matthijs S, et al. *Siderophore-mediated iron acquisition in Pseudomonas entomophila L48 and its close relative Pseudomonas putida KT2440*. 2009.

P 19459056

- Matilla MA, et al. *Temperature and pyoverdine-mediated iron acquisition control surface motility of Pseudomonas putida*. 2007.

P 17564617

- Henríquez T, Stein NV, Jung H. *PvdRT-OpmQ and MdtABC-OpmB efflux systems are involved in pyoverdine secretion in Pseudomonas putida KT2440*. 2019.

P 30346656

- Stein NV, et al. *The ABC transporter family efflux pump PvdRT-OpmQ of Pseudomonas putida KT2440: purification and initial characterization*. 2023.

P 36807028